Regeneration across earth's systems: a comprehensive solution taxonomy

Executive summary

This review catalogs **144 distinct regeneration solutions** spanning ecological restoration, circular materials, energy and water systems, atmospheric carbon removal, regenerative built environments, socio-technical frameworks, Indigenous land/sea management, biomimetic technologies, and commons governance. Three findings stand out. **First**, the deepest and most quantitatively validated regenerative solutions are *ancient and Indigenous*: Niger's farmer-managed natural regeneration (FMNR) has restored ~5–6 million ha and ~200 million trees; Worldvision West Arnhem Land Aboriginal cultural burning has cut wildfire emissions 37.7% Frontiers and generated >4.8 million carbon credits; Clean Energy Regulator Sahel zaï pits raise drought yields up to 500%; Globalearthrepairfoundation Tanzania's Sukuma *ngitili* enclosures have rehabilitated ~500,000 ha across 833 villages. **Second**, modern engineering "regeneration" (regenerative braking, self-healing concrete, RTOs, ORC waste heat) is overwhelmingly *efficiency-coded* — recovering exergy or extending product life — while true *generative* function (net primary production, soil formation, hydrological replenishment) overwhelmingly relies on living systems. **Third**, the literature is rife with overclaiming: Allan Savory's holistic grazing, the Bastin tree-restoration potential, Rodale's >100% drawdown claim, ReFi token markets, biochar permanence, and ocean alkalinity enhancement all face peer-reviewed rebuttals that should temper engineering integration.

The taxonomy is organized into the ten user-specified domains. Each entry retains canonical naming, mechanism, scale, TRL/maturity, provenance, key citations with DOIs, quantified impact, and constraints. The **gap analysis** highlights weak engineering integration in three areas: indigenous-knowledge-based MRV (measurement, reporting, verification); coupled water-carbon-soil flux modeling; and rigorous LCA of "regenerative" claims. A **brief MBSE/SysML v2 addendum** identifies which solutions have the quantifiable flows, system boundaries, and definable parts/interfaces amenable to formal model-based systems engineering — without biasing the taxonomy itself toward that lens.

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1. Biological and ecological regeneration

1.1 UN Decade Ecosystem Restoration framework

Synonyms: Generation Restoration; ecological restoration. **Description:** UN-coordinated framework (2021–2030) targeting ⩾1 billion ha under restoration. International Climate Initiat... **Mechanism:** Removal of degradation drivers (pollution, drainage, invasives) followed by biotic/abiotic restoration. **Scale:** landscape→planetary. **Maturity:** policy mainstream; underlying techniques TRL 4–9. **Origin:** UNGA Res. 73/284 (2019); SER Primer (2004). **Refs:** UNEP & FAO (2021) *Becoming #GenerationRestoration*; Gann et al. (2019) *Restoration Ecology* 27(S1) DOI 10.1111/rec.13035. **Impact:** could deliver ~⅓ of mitigation needed by 2030; Wikipedia prevent

~60% of expected extinctions. **Critique:** "restoration washing" via plantation conflation (Lewis et al. 2019, *Nature*); top-down framings marginalize Indigenous tenure.

1.2 Rewilding (passive, trophic, Pleistocene, European)

Description: Restoration through reduced human control, often reintroducing keystone fauna ("3 Cs" — cores, corridors, carnivores). Society for Conservation Bi… **Scale:** regional→continental (Y2Y, Rewilding Europe). **Maturity:** piloted–mainstream; trophic rewilding TRL 5–7. **Origin:** Foreman 1990; Soulé & Noss 1998; Extinction Rebellion Donlan et al. 2005. **Refs:** Carver et al. (2021) *Conservation Biology* 35(6) DOI 10.1111/cobi.13730; Donlan et al. (2006) *Am. Naturalist* 168(5) DOI 10.1086/508027. **Impact:** Yellowstone wolves catalyzed TABLE Debates aspen recovery; Knepp Estate biodiversity surges. **Critique:** Pleistocene rewilding ecologically dubious (Rubenstein et al. 2006); Oostvaardersplassen welfare crisis; risks of "fortress conservation" displacing rural communities.

1.3 Reforestation and afforestation

Mechanism: Photosynthetic C fixation; canopy-driven hydrology recovery. **Scale:** local→global (Bonn Challenge 350 Mha). **Maturity:** mainstream; ancient practice. **Refs:** Lewis et al. (2019) *Nature* 568:25–28 DOI 10.1038/d41586-019-01026-8; Bastin et al. (2019) *Science* 365 DOI 10.1126/science.aax0848. **Impact:** Restoring 350 Mha natural forest could remove ~42 Gt C by 2100 The Conversation — **40× the storage of plantations.** ScienceDaily **Critique:** ~45% of Bonn pledges are monoculture; Inhabitat afforestation of grasslands/savannas reduces biodiversity (Veldman et al. 2015); >50% mortality common without aftercare.

1.4 Soil regeneration / soil health restoration

Mechanism: rhizosphere photosynthate, mycorrhizal aggregation, glomalin, no-till. **Refs:** Lal (2004) *Science* 304:1623–1627 DOI 10.1126/science.1097396; Minasny et al. (2017) *Geoderma* 292 DOI 10.1016/j.geoderma.2017.01.002. **Impact:** Realistic SOC sequestration 0.2–0.8 Mg C/ha/yr. Project Drawdown® **Critique:** SOC saturation; reversibility; deep-soil dynamics uncertain.

1.5 Coral reef restoration (MARRS, coral gardening, micro-fragmentation)

Mechanism: Reef Stars stabilize rubble; fragments outplanted Mars Coral from nurseries. **Maturity:** coral gardening TRL 8; MARRS TRL 7–8; assisted evolution TRL 3–5. **Origin:** Rinkevich 1995 (Israel); Mars Inc. + Indonesian villages 2011. Mars, Incorporated **Refs:** Rinkevich (2005) *ES&T* 39(12) DOI 10.1021/es0482583; Lamont et al. (2022) *Marine Policy* 143:105199. **Impact:** >60,000 Reef Stars across 10 countries; Mars, Incorporated coral cover 5– Mars, Incorporated 10%→55–60% in 2 yrs. **Critique:** US$0.4–4M/ha; doesn't address warming; clonal bottlenecks.

1.6 Watershed restoration

Refs: Mayer et al. (2007) *J. Environ. Quality* 36(4) DOI 10.2134/jeq2006.0462; Stutter et al. (2019) *J. Environ. Quality* 48 DOI 10.2134/jeq2019.01.0020. **Impact:** Riparian buffers remove 50–95% of nitrate at >10 m width; Elwha removals freed 21 Mm^3 sediment. **Critique:** Legacy pollution decades-long; private-land cooperation difficult.

1.7 Regenerative agriculture (Rodale, Gabe Brown five principles)

Refs: Moyer et al. (2020) Rodale white paper; Brown (2018) *Dirt to Soil* ISBN 9781603587631; LaCanne & Lundgren (2018) *PeerJ* 6:e4428 DOI 10.7717/peerj.4428. **Impact:** Brown's Ranch SOM 1.9%→6.1%; California State University, ... infiltration 0.5→30 in/hr. **Critique:** Rodale's claim that global adoption could draw down >100% of annual CO_2 emissions criticized by Searchinger et al. (2018, *Nature*); definition not standardized.

1.8 Agroforestry (alley cropping, silvopasture, homegardens)

Refs: Nair et al. (2009) *J. Plant Nutr. Soil Sci.* 172 DOI 10.1002/jpln.200800030; Shi et al. (2018) *Land Degrad. Develop.* 29 DOI 10.1002/ldr.3136. **Impact:** SOC ~126 Mg C/ha to 1 m, +19% vs cropland; Wiley Online Library tropical homegardens up to 2.8 Mg C/ha/yr. Royal Meteorological Society **Critique:** highly variable; light competition can reduce crop yields.

1.9 Permaculture (Mollison & Holmgren)

Refs: Mollison (1988) *Permaculture: A Designer's Manual* ISBN 0908228015; DOKUMEN.PUB Holmgren (2002) *Permaculture: Principles and Pathways*; Krebs & Bach (2018) *Sustainability* 10:3218 DOI 10.3390/su10093218. **Critique:** Largely outside peer-reviewed agronomy; Wikipedia lower per-area yields at commercial scale.

1.10 Holistic planned grazing (Savory)

Refs: Savory & Butterfield (2016) *Holistic Management* 3rd ed.; Briske et al. (2011) *Rangeland Ecol. Mgmt.* 64; Hawkins et al. (2022) *Afr. J. Range Forage Sci.* 39. **Critique:** Multiple meta-analyses find no consistent superiority over continuous grazing; Wikipedia Savory's TED claim of reversing climate change is unsupported (FCRN *Grazed and Confused* 2017). Surge

1.11 Mycoremediation (Stamets)

Refs: Stamets (2005) *Mycelium Running* ISBN 9781580085793; Rhodes (2014) *Chem. Spec. Bioavailability* 26 DOI 10.3184/095422914X14047407349335. **Impact:** *Pleurotus* reduced 20,000 ppm diesel to <200 ppm in 8–9 wks (Battelle pilot). **Critique:** mostly lab-scale; field replicability poor; metal-laden mushrooms exit food chain.

1.12 Phytoremediation

Refs: Salt, Smith & Raskin (1998) *Annu. Rev. Plant Physiol.* 49 DOI 10.1146/annurev.arplant.49.1.643; Yan et al. (2020) *Front. Plant Sci.* 11:359 DOI 10.3389/fpls.2020.00359. **Impact:** *Pteris vittata* accumulates 3,280–4,980 ppm As. PubMed Central **Critique:** slow (5–20 yr); root depth limits.

1.13 Assisted natural regeneration (ANR) / FMNR

Largely covered in domain 8 under *Sahel re-greening*. **Refs:** FAO (2019) ANR manual; Shono, Cadaweng & Durst (2007) *Restoration Ecology* 15 DOI 10.1111/j.1526-100X.2007.00274.x. Cost ~⅓ of plantations. Anralliance

1.14 Kelp forest restoration

Refs: Eger et al. (2022) *Nat. Commun.* 13:1965 DOI 10.1038/s41467-022-29521-z; Eger et al. (2022) *Biol. Reviews* 97 DOI 10.1111/brv.12850. **Impact:** 259 projects since 1957 CA across 16 countries; ecosystem services ~US$500B/yr. **Critique:** marine heatwaves negate gains; California lost >95% of bull kelp 2014–19. CA

1.15 Peatland restoration

Refs: Joosten, Tanneberger & Moen (2017) *Mires and Peatlands of Europe*; Günther et al. (2020) *Nat. Commun.* 11:1644 DOI 10.1038/s41467-020-15499-z. **Impact:** drained peats emit 5–40 t CO_2e/ha/yr; rewetting cuts 5–30 t CO_2e/ha/yr. **Critique:** transient CH_4 rise; centuries to restore peat-formation.

1.16 Wetland restoration (Ramsar)

Refs: Moreno-Mateos et al. (2012) *PLoS Biology* 10:e1001247 DOI 10.1371/journal.pbio.1001247; Macreadie et al. (2021) *Nat. Rev. Earth Environ.* 2 DOI 10.1038/s43017-021-00224-1. **Impact:** restored wetlands recover only ~74% of biodiversity, ~77% of biogeochemical functions even after 50–100 yrs.

1.17 Biodiversity / wildlife corridors

Refs: Gilbert-Norton et al. (2010) *Conservation Biology* 24 DOI 10.1111/j.1523-1739.2010.01450.x; Hilty et al. (2020) IUCN Connectivity Guidelines ISBN 9782831720289. **Impact:** corridors increase movement ~50% on average; PubMed University of Colorado Boul... Banff overpasses cut wildlife-vehicle collisions ~80%.

1.18 The Land Institute perennial grain agriculture

Refs: Glover et al. (2010) *Science* 328 DOI 10.1126/science.1188761; DeHaan et al. (2016) *Crop Sci.* 56 DOI 10.2135/cropsci2015.06.0356; Zhang et al. (2023) *Nat. Sustainability* 6:28–38 DOI 10.1038/s41893-022-00997-3. **Impact:** Kernza absorbs up to 96% more nitrate than corn-soy; The Land Institute perennial rice (PR23) maintained yields over 8 harvests in China. **Critique:** yields still 1/3–1/10 of annual wheat. Yale e360

1.19 Natural farming (Fukuoka)

Refs: Fukuoka (1978) *The One-Straw Revolution* ISBN 9781590173138; Fukuoka (2012) *Sowing Seeds in the Desert*. One-Straw Revolution Influenced India's "Zero Budget Natural Farming" — adopted by Andhra Pradesh for ~6 million farmers / 8 Mha.

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2. Material and circular-economy regeneration

2.1 Ellen MacArthur Foundation circular economy framework

Refs: EMF (2013) *Towards the Circular Economy*; Kirchherr, Reike & Hekkert (2017) *RCR* 127

DOI 10.1016/j.resconrec.2017.09.005. **Impact:** projected ~US$1 trillion/yr materials savings; 48% EU CO_2 cut by 2030. **Critique:** 114 competing definitions; weak engagement with social equity (Corvellec et al. 2022).

2.2 Performance economy (Stahel)

Refs: Stahel (2010) *The Performance Economy* ISBN 978-0-230-58466-2; Stahel (2016) *Nature* 531:435–438 DOI 10.1038/531435a. ~65% CO_2 reduction modeled across 12 European economies if shifted to CE. Sustainabilitymag

2.3 Cradle to Cradle (McDonough & Braungart)

Refs: McDonough & Braungart (2002) Marmot ISBN 978-0865475878; Braungart, McDonough & Bollinger (2007) *J. Cleaner Prod.* 15 DOI 10.1016/j.jclepro.2006.08.003. **Critique:** Bjørn & Hauschild (2013) DOI 10.1111/j.1530-9290.2012.00520.x — ignores planetary carrying capacity, lacks LCA grounding.

2.4 Industrial symbiosis (Kalundborg)

Refs: Jacobsen (2006) *J. Industrial Ecology* 10 DOI 10.1162/108819806775545411; Chertow (2000) *Annu. Rev. Energy Env.* 25. **Impact:** ~635,000 t CO_2/yr avoided; Council Fire LLC 4 Mm^3/yr water saved; €24M/yr net benefit. **Critique:** organically evolved over 50+ yrs; planned eco-parks frequently fail.

2.5 Remanufacturing

Refs: Steinhilper (1998); Jena, Mishra & Moharana (2020) *RCR* 156:104681 DOI 10.1016/j.resconrec.2019.104681. **Impact:** ~85% energy savings vs new manufacture; up to 53% CO_2 reduction. US reman ~US$43B, 180,000 jobs.

2.6 Refurbishment

Refs: Ijomah et al. (2007) *RCIM* 23 DOI 10.1016/j.rcim.2007.02.017. Refurbished smartphones ~80% lower carbon footprint (ADEME 2022).

2.7 Upcycling

Refs: Singh, Sung, Cooper, West & Mont (2019) *RCR* 150:104439 DOI 10.1016/j.resconrec.2019.104439. **Critique:** scaling challenges; thermodynamic legitimacy contested.

2.8 Closed-loop recycling

Refs: Geyer, Jambeck & Law (2017) *Sci. Adv.* 3:e1700782 DOI 10.1126/sciadv.1700782; Hopewell, Dvorak & Kosior (2009) *Phil. Trans. R. Soc. B* 364 DOI 10.1098/rstb.2008.0311. **Impact:** Aluminum saves ~95% energy; only ~9% of all plastics ever recycled.

2.9 Urban mining

Refs: Forti, Baldé, Kuehr & Bel (2024) *Global E-Waste Monitor 2024*; Zeng, Mathews & Li (2018)

ES&T 52 DOI 10.1021/acs.est.7b04909. **Impact:** 62 Mt e-waste/yr (~US$62B); only 22% formally collected; 1 t mobile phones = 300 g gold (vs. 5 g/t ore).

2.10 Design for disassembly (DfD)

Refs: Crowther (2005); Sanchez & Haas (2018) *J. Cleaner Prod.* 183 DOI 10.1016/j.jclepro.2018.02.201. **Impact:** can cut embodied C 30–80% across multiple lifecycles.

2.11 Biodegradable polymers (PLA, PHA)

Refs: Rosenboom, Langer & Traverso (2022) *Nat. Rev. Materials* 7 DOI 10.1038/s41578-021-00407-8. **Impact:** PLA cradle-to-gate ~0.5–1.6 kg CO_2e/kg vs ~3.5 for PET. **Critique:** PLA needs industrial composting; recent ecotox findings show PLA/PHA microplastics still induce oxidative stress.

2.12 Mycelium materials (Ecovative, MycoWorks, Mogu)

Refs: Jones et al. (2020) *Materials & Design* 187:108397 DOI 10.1016/j.matdes.2019.108397. ~50% lower CO_2 than EPS packaging; home-compostable in 30–45 days.

2.13 Bio-based composites (hemp/flax/bamboo)

Refs: Pickering, Efendy & Le (2016) *Composites A* 83 DOI 10.1016/j.compositesa.2015.08.038; Bourmaud et al. (2018) *Prog. Materials Sci.* 97 DOI 10.1016/j.pmatsci.2018.05.005. ~40% lighter than glass fiber at comparable specific stiffness.

2.14 Waste-to-energy & resource systems

Refs: Brunner & Rechberger (2015) *Waste Mgmt.* 37 DOI 10.1016/j.wasman.2014.02.003; Malinauskaite et al. (2017) *Energy* 141 DOI 10.1016/j.energy.2017.11.128. **Critique:** lock-in effect; Sweden repealed its WtE tax in 2022. ScienceDirect

2.15 Product-as-a-Service (PaaS)

Refs: Tukker (2004) *Bus. Strategy Env.* 13 DOI 10.1002/bse.414; Bocken et al. (2016) *J. Industrial Prod. Eng.* 33 DOI 10.1080/21681015.2016.1172124. Springer **Impact:** JRC 2025 vacuum case: 100% collection (vs 41%), 90% reuse (vs 0%). **Critique:** Tukker (2015) review found PSS often fails to reduce environmental impact.

2.16 Repair economies / right to repair

Refs: EU Directive 2024/1799 (in force 30 July 2024); Svensson-Hoglund et al. (2021) *J. Cleaner Prod.* 288:125488. **Impact:** EU premature disposal generates 261 Mt CO_2e/yr; doubling smartphone life cuts lifecycle GHG ~30%.

2.17 Chemical recycling / depolymerization

Refs: Vollmer et al. (2020) *Angew. Chem. Int. Ed.* 59 DOI 10.1002/anie.201915651; Lopez-Davila et al. (2020) *Waste Mgmt.* 105 DOI 10.1016/j.wasman.2020.01.038. **Critique:** most pyrolysis output is fuel not feedstock (CIEL 2020; Zero Waste Europe 2023); enables continued virgin production.

2.18 Battery recycling and second-life batteries

Refs: Harper et al. (2019) *Nature* 575 DOI 10.1038/s41586-019-1682-5; Hua et al. (2024) *Nat. Commun.* 15:4179 DOI 10.1038/s41467-024-48554-0. **Impact:** EU Reg. 2023/1542 targets 90% Co/Cu/Ni/Pb recovery by 2027. Reuse + late-life recycling cuts lifecycle GHG ~15–70%.

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3. Energy regeneration and recovery

3.1 Regenerative braking / KERS

Refs: Pugi et al. (2023) *Vehicles* 5:387–403 DOI 10.3390/vehicles5020022; Boretti (2013) SAE 2013-01-2872 DOI 10.4271/2013-01-2872. Round-trip efficiency 60–70%; up to 30% urban range extension.

3.2 Ambient energy harvesting (piezo, thermo, RF)

Refs: Covaci & Gontean (2020) *Sensors* 20:3512 DOI 10.3390/s20123512; Ghazanfarian & Mohammadi (2021) *Actuators* 10:312 DOI 10.3390/act10120312. Power density typically µW–mW/cm^3.

3.3 Industrial waste heat recovery

Refs: US DOE/BCS (2008); Forman et al. (2016) *RSER* 57 DOI 10.1016/j.rser.2015.12.192. **Impact:** 20–50% of industrial energy lost as waste heat; ORC EU potential 6.6 GW_e.

3.4 District heating from waste streams

Refs: Levihn (2017) *Energy* 137 DOI 10.1016/j.energy.2017.01.118; Wahlroos et al. (2018) *RSER* 82 DOI 10.1016/j.rser.2017.10.058. Stockholm Open DH heats ~30,000 apartments via >30 data centers.

3.5 Biogas via anaerobic digestion

Refs: Khan et al. (2024) *Energy Conv. Mgmt.: X* 22:100533 DOI 10.1016/j.ecmx.2024.100533; Kapoor et al. (2021) *RSER* 138:110628 DOI 10.1016/j.rser.2020.110628. 100 t/d food waste = 800–1,400 homes powered.

3.6 Regenerative storage (V2G, pumped hydro, gravity)

Refs: Sovacool et al. (2018) *Annu. Rev. Env. Res.* 43 DOI 10.1146/annurev-environ-030117-020220; Rehman, Al-Hadhrami & Alam (2015) *RSER* 44 DOI 10.1016/j.rser.2014.12.040. PHS = >90% of global grid storage capacity; round-trip 70–85%.

3.7 Regenerative thermal oxidizer (RTO)

Refs: US EPA (2003) EPA-452/F-03-021; Choi & Yi (2000) *Chem. Eng. J.* 76 DOI 10.1016/S1385-8947(99)00120-7. VOC destruction 98–99.5%; thermal efficiency 95–97%.

3.8 Biohydrogen via dark fermentation

Refs: Rittmann & Herwig (2012) *Microbial Cell Fact.* 11:115 DOI 10.1186/1475-2859-11-115; Soares et al. (2024) *Methane* 3 DOI 10.3390/methane3030029. Theoretical max 4 mol H_2/mol glucose; practical 1–2.5.

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4. Built environment and infrastructure

4.1 Regenerative development & design (Regenesis Group)

Refs: Mang & Reed (2012) *Building Research & Information* 40 DOI 10.1080/09613218.2012.621341; Reed (2007) *BRI* 35 DOI 10.1080/09613210701475753; Mang, Haggard & Regenesis (2016) Wiley ISBN 978-1-118-97286-1. Builds on Lyle (1994).

4.2 Living Building Challenge (ILFI)

Refs: ILFI *LBC 4.1 Standard* (2024); Eisenberg & McLennan (2011) *BRI* 39 DOI 10.1080/09613218.2011.554629. Bullitt Center: EUI ~10 kBtu/ft^2/yr — ~80% below code.

4.3 Biomimetic buildings (Eastgate Centre)

Refs: Turner & Soar (2008); Benyus (1997) *Biomimicry*. **Impact:** ~35% total-energy reduction vs comparable Harare buildings (rigorous figure; popular "90%" overstated).

4.4 Green infrastructure (LID/SUDS/WSUD)

Refs: US EPA (2025) *GI Strategic Agenda 2035*; Benedict & McMahon (2006) ISBN 978-1-55963-558-0. Bioretention cuts runoff 40–80%, peaks 50–90%.

4.5 Sponge cities (Yu Kongjian, China)

Refs: Yu (2015) *City Planning Review* 39; Chan et al. (2018) *Land Use Policy* 76 DOI 10.1016/j.landusepol.2018.03.005. **Critique:** 19/30 pilot cities still flooded post-implementation (e.g., Zhengzhou 2021).

4.6 Regenerative urbanism

Refs: Girardet (2015) *Creating Regenerative Cities* DOI 10.4324/9781315764375; du Plessis (2012) *BRI* 40 DOI 10.1080/09613218.2012.628548.

4.7 Stream daylighting (Cheonggyecheon)

Refs: Lee & Anderson (2013) *J. Urban Tech.* 20 DOI 10.1080/10630732.2013.855511; Khirfan (2025) DOI 10.1080/09640568.2025.2526614. **Impact:** biodiversity +639%; $PM_{2.5}$ −35%; ambient T −3.6°C. **Critique:** pumped Han River water = "engineered fish tank"; gentrification.

4.8 IUCN Nature-based Solutions

Refs: IUCN (2020) *Global Standard for NbS* DOI 10.2305/IUCN.CH.2020.08.en; Cohen-Shacham et al. (2019) *Env. Sci. Policy* 98 DOI 10.1016/j.envsci.2019.04.014. Could provide ~37% of mitigation needed for Paris.

4.9 Biophilic design (Kellert; Browning)

Refs: Browning, Ryan & Clancy (2014/2024) *14 Patterns of Biophilic Design*; Kellert, Heerwagen & Mador (2008) Wiley ISBN 978-0-470-16334-4. Productivity gains ~6–15%; Ulrich (1984) *Science* — view of nature reduced post-op recovery.

4.10 Passivhaus with regenerative components

Refs: Feist et al. (2005) *Energy & Buildings* 37 DOI 10.1016/j.enbuild.2005.06.020; PHI standards v.10c (2023). Verified consumption ~13 kWh/m²/yr — ~90% less heating than typical German stock.

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5. Water systems regeneration

5.1 Managed aquifer recharge (MAR)

Refs: Dillon et al. (2019) *Hydrogeology Journal* 27 DOI 10.1007/s10040-018-1841-z. Global MAR ≈ 10 km³/yr, growing ~5%/yr.

5.2 Constructed wetlands

Refs: Vymazal (2011) *ES&T* 45 DOI 10.1021/es101403q; Vymazal (2009) *Ecological Engineering* 35 DOI 10.1016/j.ecoleng.2008.08.016. BOD removal 80–95%; >90% Cu/Pb/Zn removal.

5.3 Riparian restoration

Refs: Dosskey et al. (2010) *JAWRA* 46 DOI 10.1111/j.1752-1688.2010.00419.x. N removal up to ~90% in saturated buffers.

5.4 Rainwater harvesting (RWH)

Refs: Campisano et al. (2017) *Water Research* 115 DOI 10.1016/j.watres.2017.02.056; Liu et al. (2025) *Nat. Commun.* DOI 10.1038/s41467-025-66429-w. Could supply safe drinking water to 0.45–2.08 billion people.

5.5 Greywater/blackwater regeneration

Refs: Khajvand et al. (2022) *Water Sci. Tech.* 86 DOI 10.2166/wst.2022.226. SF Salesforce Tower saves 7.8M gal/yr.

5.6 Desalination with brine valorization

Refs: Panagopoulos & Giannika (2022) *J. Env. Mgmt.* 324:116239 DOI 10.1016/j.jenvman.2022.116239; Jones et al. (2019) *STOTEN* 657 DOI 10.1016/j.scitotenv.2018.12.076. Global brine ≈ 142 Mm^3/d (>50% MENA).

5.7 River daylighting

Refs: Wild et al. (2011) *Water and Environment J.* 25 DOI 10.1111/j.1747-6593.2010.00236.x. Tibbetts Brook (NYC, $133M) will divert ~220 Mgal/yr from CSO.

5.8 Beaver Dam Analogues (BDAs)

Refs: Wheaton et al. (2019) *LTPBR Design Manual* DOI 10.13140/RG.2.2.19590.63049; Bouwes et al. (2016) *Sci. Reports* 6:28581 DOI 10.1038/srep28581. Bridge Creek: smolt survival +52%; cost ~$10K/stream-mile vs $1M conventional.

5.9 Leaky weirs / leaky dams

Refs: van Leeuwen et al. (2024) *J. Hydrology* 628:130449 DOI 10.1016/j.jhydrol.2023.130449. ~10% peak reduction for events up to 1-yr return.

5.10 Slow Water movement (Gies)

Refs: Gies (2022) *Water Always Wins* ISBN 978-0226719603; Gies (2018) *Nature* 564:162 DOI 10.1038/d41586-018-07630-4.

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6. Atmosphere and carbon

6.1 Biochar

Refs: Woolf et al. (2010) *Nat. Commun.* 1:56 DOI 10.1038/ncomms1053; Lehmann et al. (2021) *Nat. Geosci.* 14 DOI 10.1038/s41561-021-00852-8; Cowie et al. (2025) *Commun. Earth Env.* DOI 10.1038/s43247-025-02228-x. Sustainable potential 0.5–2 $GtCO_2e$/yr.

6.2 Enhanced rock weathering (ERW)

Refs: Beerling et al. (2020) *Nature* 583 DOI 10.1038/s41586-020-2448-9; Beerling et al. (2024) *PNAS* 121 DOI 10.1073/pnas.2319436121; Reinhard & Planavsky (2025) *Nat. Rev. Earth Env.* DOI 10.1038/s43017-025-00713-7. CDR potential 0.5–2 $GtCO_2$/yr by 2050; cumulative 10.5 tCO_2/ha after 4 annual basalt applications.

6.3 Ocean alkalinity enhancement (OAE)

Refs: Renforth & Henderson (2017) *Rev. Geophys.* 55 DOI 10.1002/2016RG000533; NASEM (2022) *Ocean CDR*; Bach et al. (2019) *Front. Climate* 1:7 DOI 10.3389/fclim.2019.00007. **Critique:** trace metal release harms picophytoplankton; "white ocean" feedback risks.

6.4 Direct air capture with mineralization

Refs: Deutz & Bardow (2021) *Nat. Energy* 6 DOI 10.1038/s41560-020-00771-9; Matter et al. (2016) *Science* 352 DOI 10.1126/science.aad8132; NASEM (2019) *Negative Emissions Technologies*. Energy ~2,000–3,000 kWh/tCO_2; current cost $600–1,000/t.

6.5 Kelp sinking / macroalgae CDR

Refs: Krause-Jensen & Duarte (2016) *Nat. Geosci.* 9 DOI 10.1038/ngeo2790; Ricart et al. (2022) *ERL* 17:081003 DOI 10.1088/1748-9326/ac82ff; Bauer et al. (2026) *Commun. Earth Env.* DOI 10.1038/s43247-026-03342-0. **Critique:** Bauer et al. found >90% of sunken kelp biomass decomposes <100 days at depth.

6.6 Marine permaculture (von Herzen / Climate Foundation)

Refs: Hawken (2017) *Drawdown*; Climate Foundation. Demonstrated coral bleaching reversal in American Samoa via SWAC + upwelling. Largely demonstration scale.

6.7 Blue carbon ecosystems

Refs: Macreadie et al. (2021) *Nat. Rev. Earth Env.* 2 DOI 10.1038/s43017-021-00224-1; Alongi (2020) *Sci.* 2:67 DOI 10.3390/sci2030067. Mangrove stock ~739 Mg C/ha.

6.8 Methane reduction in ruminants (Asparagopsis, Bovaer/3-NOP)

Refs: Kinley et al. (2020) *J. Cleaner Prod.* 259:120836 DOI 10.1016/j.jclepro.2020.120836; Roque et al. (2021) *PLOS ONE* 16:e0247820 DOI 10.1371/journal.pone.0247820; EU Reg. 2022/565. Asparagopsis cuts CH_4 80–98%; 3-NOP 30–40%.

6.9 Cover-crop / conservation agriculture soil C

Refs: Poeplau & Don (2015) *AEE* 200 DOI 10.1016/j.agee.2014.10.024; Bai et al. (2019) *Glob. Change Biol.* 25 DOI 10.1111/gcb.14658. Mean SOC sequestration 0.32 ± 0.08 tC/ha/yr. **Critique:** Chaplot & Smith (2023) — most claims based on shallow sampling without equivalent soil mass correction.

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7. Socio-technical and systemic frameworks

7.1 Regenerative Design (Lyle 1994)

Lyle, J.T. (1994) *Regenerative Design for Sustainable Development*. Wiley. ISBN 978-0471178439. Foundational citation underlying Regenesis/Mang-Reed and Wahl traditions.

7.2 Biomimicry (Benyus 1997 / Biomimicry 3.8)

Benyus (1997) ISBN 0-688-13691-X. AskNature.org database. Biomimicry Global Network 12,576 practitioners across 21 countries.

7.3 Cradle to Cradle design (covered in §2.3)

7.4 Doughnut Economics (Raworth)

Raworth (2017) ISBN 978-1-84794-137-4. Adopted by Amsterdam (April 2020), Copenhagen, Brussels, Dunedin, Nanaimo; ~50+ cities engaging via DEAL.

7.5 Capra & Luisi *Systems View of Life*

(2014) Cambridge UP DOI 10.1017/CBO9780511895555. Foundational meta-paradigm.

7.6 Regenerative Development (Mang & Reed / Regenesis)

Mang, Haggard & Regenesis (2016) Wiley ISBN 978-1-118-97286-1. Underpins LENSES and BNIM REGEN tools.

7.7 Designing Regenerative Cultures (Wahl 2016)

ISBN 978-1-909470-77-4. Bridges Capra, biomimicry, permaculture, regenerative design.

7.8 Regenerative Leadership (Hutchins & Storm 2019)

ISBN 978-1-78324-119-4. Future Fit Leadership Academy.

7.9 Theory U / Presencing (Scharmer)

Scharmer (2007/2016) Berrett-Koehler. u.lab MOOC reaches >140,000 users in 185 countries.

7.10 Permaculture design (covered in §1.9)

7.11 Circular business model archetypes (Bocken et al.)

Bocken, Short, Rana & Evans (2014) *J. Cleaner Prod.* 65 DOI 10.1016/j.jclepro.2013.11.039. Eight archetypes across technological, social, organizational dimensions.

7.12 Regenerative Finance (ReFi)

Fullerton (2015) *Regenerative Capitalism* (Capital Institute); Bennett et al. (2023) *Front. Blockchain* 6:1165133 DOI 10.3389/fbloc.2023.1165133. **Critique:** Toucan/Klima crashed ~99.99% from peak; Verra paused tokenization (May 2022); ~half of self-described ReFi projects not truly regenerative.

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8. Indigenous and traditional regenerative practices

These are first-class regenerative solutions, not appendix material. Where peoples and language names are given, they are precise; specific land/sea practices listed below are accompanied by quantified outcomes wherever documented.

Africa

8.1 Zaï pits (Mossi/Sawadogo, Burkina Faso)

Refs: Reij, Tappan & Smale (2009) IFPRI DP 00914; Right Livelihood Foundation (2018) Sawadogo profile. Drought yields up to **+500%**; tens of thousands of ha restored; key contribution to ~6 million ha Sahel re-greening.

8.2 Stone bunds / cordons pierreux

Refs: Critchley & Reij (1989) World Bank Tech Paper 91; Reij/Tappan/Smale 2009. Yields +40–100%; water tables rose >5 m in some treated villages.

8.3 Farmer-Managed Natural Regeneration (FMNR — Rinaudo & Hausa/Zarma farmers)

Refs: Rinaudo (2022) *The Forest Underground*; Tougiani, Guero & Rinaudo (2009) *GeoJournal* 74 DOI 10.1007/s10708-008-9228-7. **Impact:** ~5–6 million ha restored in Niger; ~200 million additional trees; ~2.5 million people benefited; ~24 million ha across 11 nations. Cost ~$20–50/ha vs ~$8,000/ha conventional planting. Niger 1993 Rural Code (transferring tree-tenure to farmers) was the pivotal enabler.

8.4 Ethiopian Orthodox church forests

Refs: Wassie, Sterck & Bongers (2010) *J. Veg. Sci.* 21 DOI 10.1111/j.1654-1103.2010.01202.x; Cardelús, Lowman & Eshete (2012) *Science* 335 DOI 10.1126/science.335.6071.915-a. Principal remnant of native afromontane forest in Amhara; ⩾20 communities have erected protective walls.

8.5 Ngitili enclosures (Sukuma, Tanzania)

Refs: Barrow & Mlenge (2003); Monela et al. (2005) IUCN/MNRT; Pye-Smith (2010) ICRAF. **Impact:** ~378,000–500,000 ha restored across 833 villages; >2.5 million beneficiaries; ~US$14/person/month value; ~23.21 Mt C sequestered over 25 yrs (Equator Initiative Award 2002).

8.6 Mafia Island traditional fishing / MIMP (Tanzania)

Refs: Walley (2004) *Rough Waters*; Benjaminsen & Bryceson (2012) *J. Peasant Studies* 39 DOI 10.1080/03066150.2012.667405. 400+ fish species; reefs among healthiest in Indian Ocean. **Critique:** social-science literature flags livelihood displacement.

8.7 Konso terraced agriculture (Ethiopia)

Refs: UNESCO (2011) WHL #1333; Watson (2009) *Living Terraces in Ethiopia* ISBN 9781847010179. ⩾21 generations of continuous practice (~400+ years); UNESCO World Heritage.

8.8 Maasai/East African pastoralist rotational grazing

Refs: Western, Russell & Cuthill (2009) *PLoS ONE* 4:e6140 DOI 10.1371/journal.pone.0006140;

Schuette, Creel & Christianson (2016) *J. Appl. Ecol.* 53 DOI 10.1111/1365-2664.12610. **Impact:** Mara conservancies hold ~84% of ecosystem wildlife outside the formal reserve.

Oceania

8.9 Aboriginal "cool burning" / WALFA

Refs: Russell-Smith et al. (2013) *Front. Ecol. Env.* 11 DOI 10.1890/120251; Bliege Bird et al. (2008) *PNAS* 105 DOI 10.1073/pnas.0804757105; Bird et al. (2016) *Curr. Anthropol.* 57:S65 DOI 10.1086/685763. **Impact:** WALFA cut wildfire GHG emissions **37.7%** vs 10-yr baseline; ALFA projects manage >80,000 km^2 and have generated **>4.8 million ACCUs**.

8.10 Kaitiakitanga (Māori, Aotearoa)

Refs: Royal (2007) *Te Ara*; Kawharu (2000) *J. Polynesian Soc.* 109. Statutorily recognized in NZ Resource Management Act 1991 §7(a); Te Awa Tupua (Whanganui River) Act 2017 granted legal personhood.

8.11 Pacific Islander multi-story agroforestry

Refs: Raynor & Fownes (1991) *Agroforestry Systems* 16 DOI 10.1007/BF00129745; Clarke & Thaman (1993) UN Univ. Press ISBN 9280808249. Pohnpei: 161 plant species in 10 ha; 28 breadfruit + 38 yam cultivars.

8.12 Ahupua'a — Hawaiian watershed management

Refs: Winter, Lincoln & Berkes (2018) *Sustainability* 10:3294 DOI 10.3390/su10093294; Gon & Winter (2019) *Am. Scientist* 107 DOI 10.1511/2019.107.4.232.

8.13 Loko i'a (Hawaiian fishponds) and lo'i kalo

Refs: Costa-Pierce (1987) *BioScience* 37 DOI 10.2307/1310688; Kikuchi (1976) *Science* 193 DOI 10.1126/science.193.4250.295. Pre-contact ponds ~150–300 kg fish/ha/yr; oldest dated ~1,500–1,800 BP.

8.14 Fijian iqoliqoli / tabu areas

Refs: Aswani (2005) *Rev. Fish Biol. Fisheries* 15 DOI 10.1007/s11160-005-4868-x; Jupiter & Egli (2011) *J. Marine Biol.* 2011:940765 DOI 10.1155/2011/940765. **Impact:** Ucunivanua kaikoso clam tabu — clams in plot rose from 38 (1997) to >7,000 (2004).

Americas

8.15 Three Sisters / Diohe'ko polyculture (Haudenosaunee)

Refs: Mt. Pleasant (2016) *Ethnobiology Letters* 7 DOI 10.14237/ebl.7.1.2016.721; Lewandowski (1987) *Agriculture and Human Values* 4 DOI 10.1007/BF01530644; Kimmerer (2013) *Braiding Sweetgrass*. 1 ha supports ~13 people·yr^{-1} for energy, ~15 for protein.

Refs: Donkin (1979) Univ. Arizona Press; Chepstow-Lusty (2011) *Climate of the Past* 5; Inbar & Llerena (2000) *Mtn. Res. Develop.* 20. Stable nutrient profiles for >1,000 yrs documented at Marcacocha.

8.17 Waru waru / camellones (Tiwanaku, Aymara/Quechua)

Refs: Erickson (1988) *Expedition* 30; Bandy (2005) *J. Anthropol. Archaeol.* 24. Potato yields 11,000–22,000 kg/ha — 2–4× regional dryland average.

8.18 Chinampas (Aztec/Nahua)

Refs: Ebel (2020) *HortTechnology* 30 DOI 10.21273/HORTTECH04310-19; Morehart (2012) *J. Archaeol. Sci.* 39. ~13× yield of dry-land farming on equivalent area; UNESCO World Heritage; FAO GIAHS (2018).

8.19 Milpa system (Maya/Mesoamerican)

Refs: Fonteyne et al. (2023) *Frontiers in Agronomy* 5:1115490 DOI 10.3389/fagro.2023.1115490; Falkowski et al. (2019) *Food Security* 11. Lacandon Maya milpa supplies all macronutrients + vitamins A/C, calcium, iron, zinc.

8.20 Acequia irrigation systems (Pueblo + Hispano)

Refs: Rivera (1998) Univ. New Mexico Press; Fernald et al. (2015) *HESS* 19 DOI 10.5194/hess-19-293-2015. ~700 functioning acequias in NM irrigating ~32,000 ha; 30–60% of diverted water returns as recharge.

8.21 Indigenous cultural burning (Karuk, Yurok, Anishinaabe)

Refs: Anderson (2005) *Tending the Wild*; Marks-Block, Lake, Bliege Bird & Curran (2021) *Fire Ecology* 17:6 DOI 10.1186/s42408-021-00092-6; Knight et al. (2022) *PNAS* 119 DOI 10.1073/pnas.2116264119; Larson et al. (2025) *PNAS* 122 DOI 10.1073/pnas.2500024122. 7–10× hazelnut basketry-stem production after burning; cultural burning kept Klamath forest biomass at ~half today's levels; Wisconsin/Minn. Point fire-scar record 1737–1999 shows fire-cessation correlates with 1842 forced removal.

8.22 Amazonian terra preta

Refs: Glaser et al. (2001) *Naturwissenschaften* 88; Lehmann et al. (eds.) (2003) *Amazonian Dark Earths*. SOC ~150–250 Mg C/ha to 1 m vs ~50 in adjacent oxisols (3–5×); black carbon stable >2,000 yrs.

8.23 Pacific Northwest clam gardens (Coast Salish, W̱SÁNEĆ, Heiltsuk, etc.)

Refs: Groesbeck, Rowell, Lepofsky & Salomon (2014) *PLOS ONE* 9:e91235; Smith et al. (2019) *PLOS ONE* 14:e0211194; Deur et al. (2015) *Human Ecology* 43. **Impact:** 4× butter clam, 2× littleneck densities; 1.7× faster transplanted juvenile growth; ⩾3,500 yrs old.

8.24 Satoyama landscapes (Japan)

Refs: Takeuchi (2010) *Ecol. Research* 25 DOI 10.1007/s11284-010-0745-8; Kadoya & Washitani (2011) *AEE* 140. Paddy ecosystems support >5,668 species; ~half of Japan's threatened species. UN Satoyama Initiative since 2008.

8.25 Korean maeul / village forests

Refs: Lee et al. (2021) *Forest Policy & Economics* 128:102476; Koh, Reineking, Park & Lee (2015) *J. Veg. Sci.* 26.

8.26 Subak water management (Bali)

Refs: Lansing (1987) *Am. Anthropologist* 89 DOI 10.1525/aa.1987.89.2.02a00030; Lansing & Kremer (1993) *Am. Anthropologist* 95; Lansing (2006) *Perfect Order* Princeton UP. ~1,200 active subaks managing ~80,000 ha; UNESCO World Heritage 2012; computer simulations show water-temple coordination yields Pareto-optimal water-vs-pest trade-off.

8.27 Indian sacred groves

Refs: Chandrashekara & Sankar (1998) *Forest Ecol. Mgmt.* 112; Bhagwat & Rutte (2006) *Front. Ecol. Env.* 4; Malhotra et al. (2001) INSA. ≥100,000–150,000 sacred groves nationally.

8.28 Mongolian rotational nomadic grazing

Refs: Fernández-Giménez (2000) *Ecol. Applications* 10; Reid, Fernández-Giménez & Galvin (2014) *Annu. Rev. Env. Res.* 39. ~70% of Mongolian rangeland (~110 Mha); ~70 million livestock; 1/3 of population still pastoralist.

8.29 Hima system (Arabian Peninsula)

Refs: Kilani, Serhal & Llewellyn (2007) IUCN/SPNL; Gari (2006) *Environment and History* 12 DOI 10.3197/096734006776680290; Al-Rowaily et al. (2015) *Saudi J. Biol. Sci.* 22 DOI 10.1016/j.sjbs.2014.09.016. ≥1,400 yrs documented; >31 himas covering >6% of Lebanon.

8.30 Zabo / cheo-oji system (Chakhesang Naga, NE India)

Refs: Sharma, Prasad & Sonowal (1994) ISCO; Singh, Singh & Rajkhowa (2012) ICAR-NEH. Paddy 1.95 t/ha (zabo) vs 1.0 t/ha (jhum); fish ~62 kg/ha.

8.31 Chinese rice-fish coculture (Qingtian)

Refs: Xie et al. (2011) *PNAS* 108:E1381–E1387 DOI 10.1073/pnas.1111043108; FAO GIAHS (2005, first-ever GIAHS site). Pesticide use cut 68%, fertilizer 24%, while adding ~750 kg/ha fish.

Europe & Sápmi

8.32 Spanish dehesa / Portuguese montado

Refs: Bugalho et al. (2011) *Front. Ecol. Env.* 9 DOI 10.1890/100084; Plieninger (2007) *J. Nature Conservation* 15 DOI 10.1016/j.jnc.2005.09.002. ~3 Mha; key habitat for Iberian lynx and Spanish imperial eagle.

8.33 Romanian/Carpathian traditional hay meadows

Refs: Babai & Molnár (2014) *AEE* 182 DOI 10.1016/j.agee.2013.08.018; Wilson et al. (2012) *J. Veg. Sci.* 23 — world record up to 98 vascular plant species per 16 m^2.

8.34 Sámi reindeer herding

Refs: Roturier & Roué (2009) *Forest Ecol. Mgmt.* 258 DOI 10.1016/j.foreco.2009.07.045; Eira et al. (2013) *Cold Reg. Sci. Tech.* 85 DOI 10.1016/j.coldregions.2012.09.004. Sámi siidas manage ~40% of Sweden, ~40% of Norway, ~35% of Finland.

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9. Bio-inspired / biomimetic technologies

9.1 Self-healing polymer composites (microcapsule)

White et al. (2001) *Nature* 409:794–797 DOI 10.1038/35057232. ~75–100% recovery of virgin fracture toughness.

9.2 Self-healing concrete (Jonkers — bacterial)

Jonkers et al. (2010) *Ecological Engineering* 36 DOI 10.1016/j.ecoleng.2008.12.036; Palin, Wiktor & Jonkers (2017) *Biomimetics* 2:13 DOI 10.3390/biomimetics2030013. Heals cracks up to 0.8 mm; permeability cut 95%.

9.3 Aquaporin biomimetic membranes

Tang et al. (2013) *Desalination* 308 DOI 10.1016/j.desal.2012.07.007; Azarafza et al. (2023) *Adv. Funct. Mater.* DOI 10.1002/adfm.202213326. 99.1% NaCl rejection at 7.83 L m^{-2} h^{-1} bar^{-1}.

9.4 Namib beetle-inspired water capture

Parker & Lawrence (2001) *Nature* 414:33–34 DOI 10.1038/35102108; Zhai et al. (2006) *Nano Letters* 6 DOI 10.1021/nl060644q. 2–5× fog-droplet capture vs plain mesh.

9.5 Fog harvesting nets (FogQuest, CloudFisher)

Schemenauer & Cereceda (1994) *J. Appl. Meteorology* 33 DOI 10.1175/1520-0450(1994)033<1313:APSFCF>2.0.CO;2; Klemm et al. (2012) *Ambio* 41 DOI 10.1007/s13280-012-0247-8. Boutmezguida (Morocco): ~37,000 L/fog day, 1,300 people, 16 villages.

9.6 Lotus-effect self-cleaning surfaces

Barthlott & Neinhuis (1997) *Planta* 202 DOI 10.1007/s004250050096; Barthlott, Mail & Neinhuis (2016) *Phil. Trans. R. Soc. A* 374 DOI 10.1098/rsta.2016.0191. Contact angles 150–170°.

Turner & Soar (2008); King, Ocko & Mahadevan (2015) *PNAS* 112 DOI 10.1073/pnas.1423242112. ~35% total energy reduction vs comparable buildings.

9.8 Bird-inspired aerodynamics (Shinkansen kingfisher; owl-feather pantograph)

Nakatsu (2011) *Hitachi Review*; Lilley (1998) AIAA-1998-2340 DOI 10.2514/6.1998-2340. ~15% energy reduction; tunnel-boom under 70 dB(A).

9.9 Bio-inspired adhesives — gecko & mussel

Autumn et al. (2000) *Nature* 405:681–685 DOI 10.1038/35015073; Autumn et al. (2002) *PNAS* 99 DOI 10.1073/pnas.192252799; Lee, Dellatore, Miller & Messersmith (2007) *Science* 318 DOI 10.1126/science.1147241; Lee, Lee & Messersmith (2007) *Nature* 448 DOI 10.1038/nature05968. Synthetic mimics 5–10 N/cm² shear adhesion.

9.10 Mycelium computing / fungal sensing (Adamatzky)

Adamatzky (2018) *Interface Focus* 8:20180029 DOI 10.1098/rsfs.2018.0029; Adamatzky et al. (2022) *Biosystems* 212:104588 DOI 10.1016/j.biosystems.2021.104588. Demonstrated logic gates and chemical detection at TRL 2–3.

9.11 Sharkskin-inspired riblet & antifouling surfaces

Bechert et al. (1997) *J. Fluid Mech.* 338 DOI 10.1017/S0022112096004673; Dean & Bhushan (2010) *Phil. Trans. R. Soc. A* 368 DOI 10.1098/rsta.2010.0201; Oeffner & Lauder (2012) *J. Exp. Biol.* 215 DOI 10.1242/jeb.063040. AeroSHARK ~1% fuel saving on long-haul aircraft. **Note:** Speedo Fastskin "shark-inspired" claim refuted for rigid bodies.

9.12 Spider silk biomimetic fibres (Bolt Threads, AMSilk, Spiber)

Heim, Keerl & Scheibel (2009) *Angew. Chem. Int. Ed.* 48 DOI 10.1002/anie.200803341; Andersson et al. (2017) *Nat. Chem. Biol.* 13 DOI 10.1038/nchembio.2269.

9.13 Velcro® (de Mestral; foundational biomimicry)

US Patent 3,009,235 (1961). Vincent et al. (2006) *J. R. Soc. Interface* 3 DOI 10.1098/rsif.2006.0127.

9.14 WhalePower tubercle blades

Fish & Battle (1995) *J. Morphology* 225 DOI 10.1002/jmor.1052250105; Miklosovic et al. (2004) *Phys. Fluids* 16:L39 DOI 10.1063/1.1688341. ~6% lift increase, up to 32% post-stall drag decrease; HVLS fans ~25% airflow gain.

9.15 Butterfly photonic structures (Morpho radiative cooling)

Vukusic & Sambles (2003) *Nature* 424 DOI 10.1038/nature01941; Potyrailo et al. (2007) *Nature Photonics* 1 DOI 10.1038/nphoton.2007.2; Didari & Mengüç (2018) *Sci. Reports* 8:16891 DOI 10.1038/s41598-018-35082-3. Sub-ambient cooling up to 8.45 °C.

Reece et al. (2011) *Science* 334:645–648 DOI 10.1126/science.1209816; Liu et al. (2016) *Science* 352 DOI 10.1126/science.aaf5039. Wireless solar-to-H_2 ~2.5%; bionic leaf ⩾10% solar-to-biomass.

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10. Social and cultural regeneration relevant to engineering

10.1 Commons-based governance (Ostrom 1990)

Ostrom (1990) *Governing the Commons* Cambridge UP DOI 10.1017/CBO9780511807763. Eight design principles; cornerstone of IASC and digital-commons governance.

10.2 Transition Towns (Hopkins)

Hopkins (2008) *The Transition Handbook* ISBN 978-1-900322-18-8. ~450 formal initiatives across 50+ countries.

10.3 Community Land Trusts (Swann et al. 1972; Davis 2010)

Swann et al. (1972) International Independence Institute; Davis (ed.) (2010) Lincoln Institute ISBN 978-1-55844-205-9. 260+ CLTs across 46 US states + DC + PR; expanding internationally.

10.4 Indigenous-led conservation (IPBES)

IPBES (2019) *Global Assessment Report* DOI 10.5281/zenodo.3831673. **At least 25% of global land** is traditionally owned/managed/used/occupied by Indigenous peoples; biodiversity declines less rapidly there.

10.5 Citizen science (eBird, iNaturalist)

Sullivan et al. (2009) *Biological Conservation* 142 DOI 10.1016/j.biocon.2009.05.006; Callaghan et al. (2024) *FEE* DOI 10.1002/fee.2808. iNaturalist >200M observations from ~3.3M observers; eBird >1B records; 228 of 1,355 US EISs (2012–2022) referenced citizen-science data.

10.6 Buen Vivir / Sumak Kawsay

Constitution of Ecuador (2008) Arts. 71–74 (Rights of Nature); Gudynas (2011) *Development* 54 DOI 10.1057/dev.2011.86; Acosta (2013) *El Buen Vivir* ISBN 978-84-9888-497-9. First constitutional Rights of Nature globally.

Cross-cutting analysis: recurring mechanisms, scale gaps, synergies

Mechanisms that recur across domains. Five mechanism families dominate. *Loop closure* (technical and biological cycles in C2C, industrial symbiosis, urban mining, anaerobic digestion) appears in 30+ entries. *Hydrological retention and slow release* underlies sponge cities, BDAs, leaky weirs, ahupua'a, waru waru, zaï, MAR, and constructed wetlands — a profound convergence between engineered and Indigenous practices. *Photosynthetic carbon transfer to recalcitrant pools* (biochar, terra preta, blue carbon, peatland, perennial grains) is the single largest leverage point for atmospheric drawdown. *Biological self-organization under managed disturbance* — fire stewardship, holistic grazing, ANR/FMNR, satoyama, hay meadows — recurs wherever humans co-create habitat heterogeneity rather than fence it off. *Co-evolutionary place-based design* (Regenesis, kaitiakitanga, Wahl, ahupua'a, dehesa) emerges as the meta-principle linking material and cultural regeneration.

Scale representation. Landscape and regional scales dominate (~60% of entries) because regeneration is intrinsically a flux phenomenon needing space and time. Molecular/component scale appears mainly in biomimetic materials, biodegradable polymers, and OAE chemistry. **Planetary-scale solutions are conspicuously few**: only DAC, ERW, OAE, the IUCN/UN frameworks, and Doughnut Economics genuinely operate or aspire to planetary scope. **Urban scale is over-represented in narrative but under-delivered in measurement**: regenerative urbanism, sponge cities, and biophilic design have abundant rhetoric but modest instrumented outcomes.

Synergies. Highest-leverage stacks combine *Indigenous land tenure + ANR/FMNR + biochar + agroforestry* (Sahel, Burkina Faso, Tanzania); *daylighted streams + sponge cities + BDAs* (urban water); *kelp forest restoration + marine permaculture + clam gardens + iqoliqoli tabu* (coastal seascape regeneration); *Passivhaus + Living Building Challenge + biophilic design + DfD* (building stack); *cover crops + perennial grains + holistic grazing + biochar + ERW* (regenerative cropland-rangeland stack). The most overlooked synergy is **Indigenous fire stewardship + carbon markets** (WALFA's >4.8M ACCUs is the clearest exemplar, but governance is fragile and "carbon colonialism" remains a real risk).

Gap analysis: what is missing or weakly integrated

1. **MRV for living-system regeneration is the binding constraint.** Biochar permanence, soil C accrual, ERW dissolution rates, OAE air-sea equilibration, peatland CH_4 vs CO_2 trade-offs, kelp-sinking decomposition, and FMNR additionality all suffer the same problem: outcome verification at adequate spatial-temporal resolution and cost. This is *the* engineering frontier and is largely absent from MBSE/SysML-friendly engineering literature.
2. **Coupled hydro-bio-geochemical modeling.** Integrative frameworks linking Indigenous-managed flows (acequia recharge, ahupua'a sediment-nutrient cascades, subak temple coordination) to formal hydrological/ecological models are nascent; Lansing's water-temple agent-based models are an exception, not the rule.
3. **LCA rigor for "regenerative" claims.** Bjørn & Hauschild (2013), Tukker (2015), Zink & Geyer (2017) "Circular Economy Rebound," and Chaplot & Smith (2023) collectively show that regenerative claims often fail when subjected to dynamic, system-boundary-corrected LCA. Greenwashing risk is highest for ReFi, NbS, mycelium materials, biodegradable polymers, and "regenerative" labels for monoculture plantations.

4. **Cross-system feedback under climate change.** Sea-level rise threatens coastal restoration; marine heatwaves negate kelp restoration; permafrost thaw could reverse tundra C sinks; ice-on-snow events from Arctic warming threaten Sámi pasture. Regeneration solutions are routinely modeled as static, when in reality they operate within trajectories of system collapse.
5. **Indigenous data sovereignty and FPIC.** Engineering literature continues to extract Indigenous knowledge into "biomimicry" and "nature-based solutions" frames without consent, attribution, or benefit-sharing — terra preta, MARRS, FMNR, and milpa-derived intercropping models are all flashpoints.
6. **Equity and gentrification.** Daylighted streams, urban NbS, biophilic design, and Living Buildings consistently produce green gentrification; engineering integration often omits the social-equity flux variables.
7. **Governance pathways** — particularly secured land/sea tenure (Niger 1993 Rural Code; NZ Te Awa Tupua Act; Australian Native Title; Ostrom's design principles) — are the single most decisive enabler of regenerative outcomes, yet are rarely modeled as system parameters.

Concluding synthesis

Regeneration is not one solution but a **distributed civilizational repertoire** spanning ancient Indigenous land and sea management, materials-science loop closure, ecosystem restoration, and emerging negative-emissions chemistry. The *most quantitatively impactful* solutions are also the *oldest*: FMNR (~6 Mha), zaï pits (+500% drought yields), ngitili (~500,000 ha), WALFA (37.7% emission cut), terra preta (millennial C stability), and subak (Pareto-optimal water-pest coordination over 1,000+ years). The most *over-claimed* solutions are also the most *novel*: ReFi, holistic planned grazing's climate claim, kelp-sinking permanence, OAE without ecological side-effects, and DAC as substitute for emissions reduction. A defensible engineering-policy stance is to **prioritize secured Indigenous tenure and proven low-cost living-system restoration before high-cost technical-cycle solutions**, while investing in MRV infrastructure that makes both legible to the same accounting systems.

The intellectual lineage uniting the best of these solutions runs from **Aldo Leopold (1949) → Bill Mollison & David Holmgren (1978) → Wes Jackson (1980) → John Tillman Lyle (1994) → Janine Benyus (1997) → William McDonough & Michael Braungart (2002) → Fritjof Capra & Pier Luigi Luisi (2014) → Daniel Christian Wahl (2016) → Kate Raworth (2017)**, integrated by Indigenous knowledge-holders writing in their own voices (Robin Wall Kimmerer, Melissa Nelson, Frank K. Lake, Marcia Langton, Yacouba Sawadogo, Tony Rinaudo). Engineering integration is strongest where these traditions intersect with rigorous flux quantification — and weakest where engineering attempts regeneration without them.

This addendum is *evaluative only* — the taxonomy above was constructed to be domain-neutral and complete. Solutions amenable to MBSE/SysML v2 share three properties: (a) quantifiable mass/energy/information flows, (b) clear system boundaries with definable interfaces, and (c) discrete parts whose interactions can be formalized.

Strong fit (TRL ⩾7, well-instrumented flows, modular interfaces): regenerative braking and KERS; regenerative thermal oxidizers; ORC waste-heat recovery; district heating from data centers and waste streams; anaerobic digestion biogas systems; Passivhaus envelopes with MVHR; aquaporin membrane modules; PHS, V2G, gravity storage; remanufacturing and battery-recycling lines; closed-loop industrial recycling; managed aquifer recharge; constructed wetlands; chemical-recycling depolymerization trains; DAC + mineralization plants; Kalundborg-type industrial symbiosis exchanges; AeroSHARK riblet films; Living Building Challenge buildings (fully metered).

Moderate fit (well-defined parts but fuzzier flux/control): sponge-city stormwater networks; green infrastructure (bioswales, green roofs); BDAs and leaky weirs (process-based geomorphic models exist); rice-fish coculture; biochar pyrolysis units; ERW field-application systems; mycelium materials production; second-life battery packs; product-as-a-service fleets; Hawaiian loko i'a (defined hydraulics, sluice gates); subak (Lansing's agent-based models prove tractability).

Weak fit (essential to the regenerative landscape but resistant to formal modeling): rewilding trajectories; permaculture design; regenerative leadership; kaitiakitanga; Buen Vivir; commons governance under Ostrom's principles; cultural fire stewardship as a *practice* (carbon flux is modelable; the stewardship relation is not); regenerative business models; transition towns; biophilic design experience effects; story-of-place methodology. These resist MBSE not because they are unrigorous but because they encode relational, intergenerational, and spiritual variables outside the system-engineering ontology — a reminder that the most powerful regeneration solutions integrate dimensions that engineering alone cannot represent.